

Conjugated Systems & Aromaticity

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July 2026

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1 Diel Alder Reactions

Core Concept 1.1: Cycloadditions

Cycloadditions refer to the reaction of two separate π systems to form a new ring, an interaction that involves neither a nucleophile nor an electrophile.

Cycloadditions have already been encountered in two reactions within Alkenes & Alkynes.

(a) **Ozonolysis** (Core Concept 1.18)

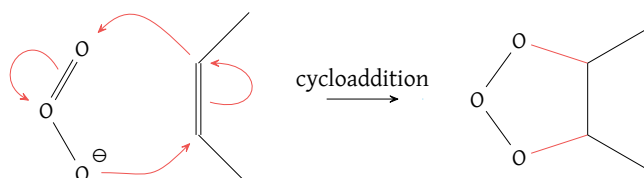


Figure 1: Ozonolysis

(b) **OsO₄ Syn-Dihydroxylation** (Core Concept 1.17)

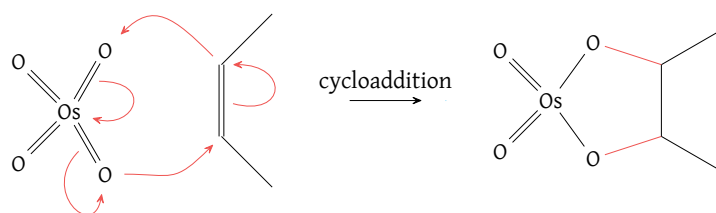


Figure 2: OsO₄ syn-dihydroxylation

Core Concept 1.2: Diel Alder Reactions

Diels-Alder reactions are a type of cycloaddition that combine a diene possessing a minimum of $4\pi e^-$ s with a dienophile possessing a minimum of $2\pi e^-$ s, forming a cyclohexene product.

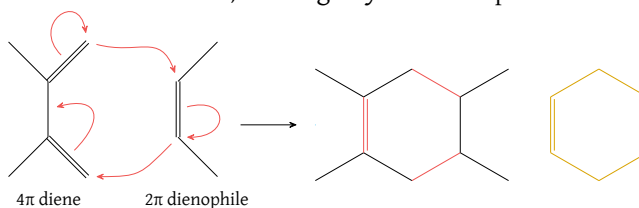


Figure 3: Diel alder reaction; combination of a minimum 4π diene wiht a minimum 2π dienophile

Consequently, diene and dienophile pairings possessing conjugated π systems larger than these stated minimums of 4π and 2π , respectively, can give rise to a substantial diversity of stereoisomeric products.

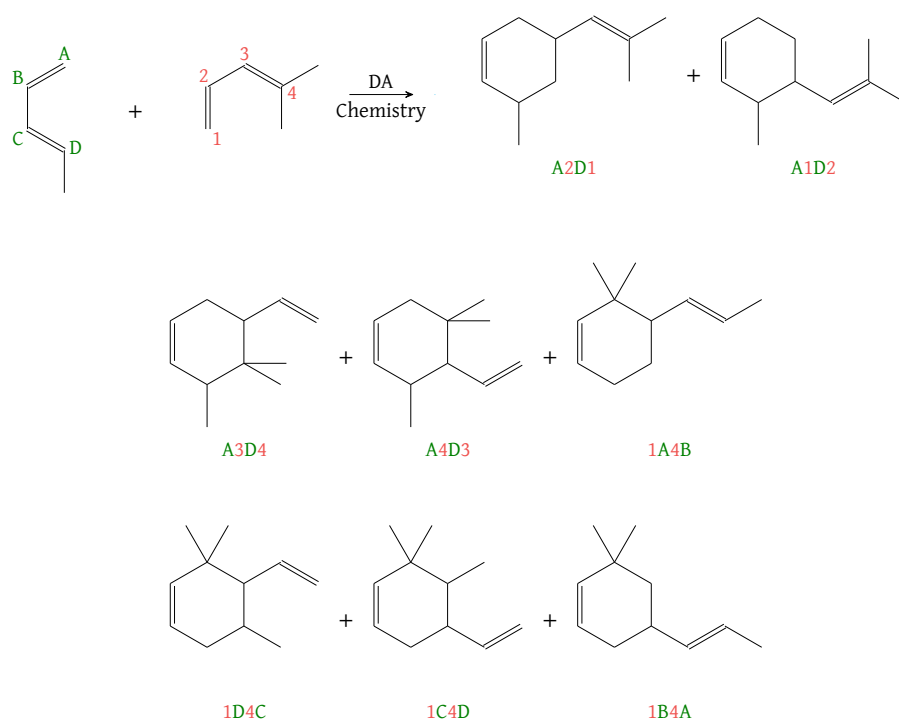
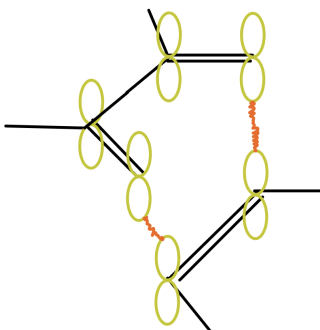


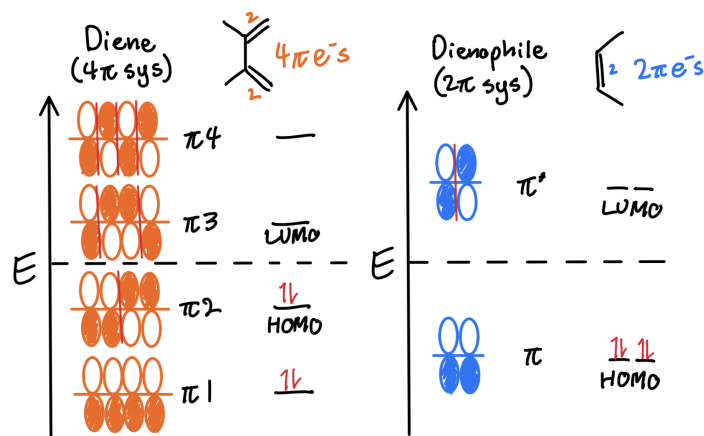
Figure 4: Diel alder reaction product diversity

Core Concept 1.3: Orbitals of Diel Alder Reactions

In Diels–Alder reactions, the π systems of the diene and dienophile approach one another in a vertical orientation, enabling them to generate new σ overlaps.

Figure 5: Diene and dienophile vertical interaction; *p orbitals overlap to form σ interactions*

Consider the energy diagrams of the respective conjugated π systems of the diene and dienophile.

Figure 6: Conjugated π system energy diagrams of a diene and dienophile

A key observation follows from examining which energy levels interact with which: the HOMO of one reagent interacts with the LUMO of the other.

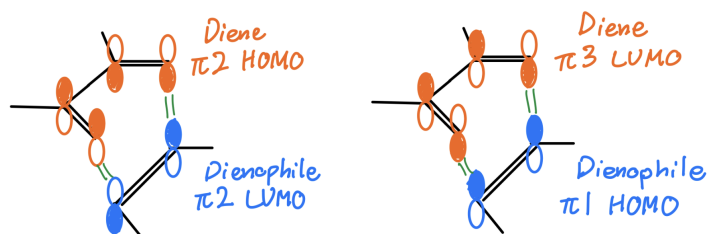


Figure 7: HOMO-LUMO interaction versions between diel alder reagents

Core Concept 1.4: Diel Alder Stereochemistry

Diels-Alder reactions can further be categorized according to the distinct stereochemical approaches the reagents may adopt.

- (a) **Top** and **bottom** Diels-Alder reactions refer, respectively, to the dienophile approaching the diene from **above** or **below** its plane.



Figure 8: Bottom (left) and top (right) attacks by the dienophile in a diel alder reaction

- (b) **Endo** and **exo** distinctions refer to whether the substituents of the dienophile are oriented **towards** or **away from** the corresponding diene.

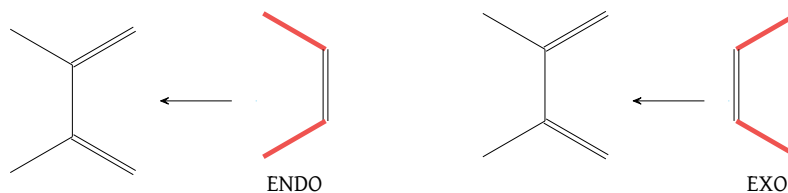


Figure 9: Endo (*left*) and exo (*right*) attack by the dienophile in a Diels-Alder reaction

It is important to note that the bonds oriented towards the diene are flipped about the dienophile's double bond, in the direction opposite to the diene's approach.

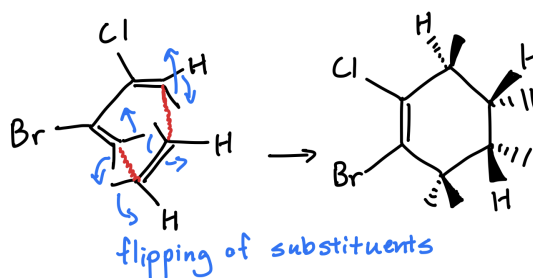


Figure 10: Flipping of dienophile substituents pointing towards the diene during a Diels-Alder reaction

In general, this flipping process can be summarized by the following outline reaction.

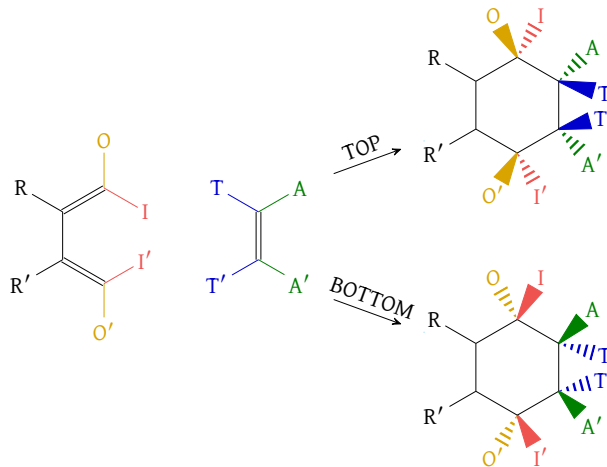


Figure 11: Diastereoselective outcomes resulting from top-face (*top*) versus bottom-face (*bottom*) attack during a cycloaddition reaction

As practice, consider the following Diels-Alder reaction, in which the objective is to predict all non-dimeric Diels-Alder products.

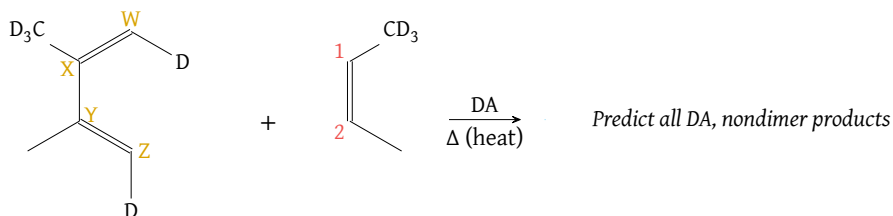


Figure 12: Example diel alder reaction

Two possible constitutional isomers can form, corresponding to the two distinct positionings the dienophile may adopt.

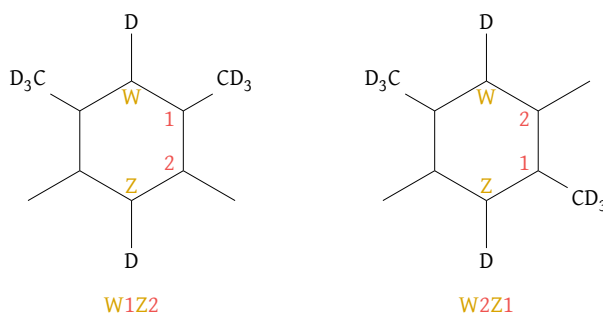


Figure 13: Isomer product possibilities of the reaction shown in Figure 12

It is also important to note that for each of these diene–dienophile arrangements, four distinct stereochemical approaches exist—endo top, endo bottom, exo top, and exo bottom—consistent with the general outline given in Figure 11.

1.1 Rates of Diel Alder Reactions

Core Concept 1.5: Diene Orientation

Diels-Alder reactions proceed more favorably when the resulting ring exhibits minimal strain. The **s-cis** diene orientation satisfies this condition more effectively than the **s-trans** orientation, as illustrated below. The "s" in s-cis and s-trans specifically denotes orientation about a single bond, distinguishing this usage from the conventional application of cis/trans nomenclature to alkenes, where the double bond serves as the reference axis.

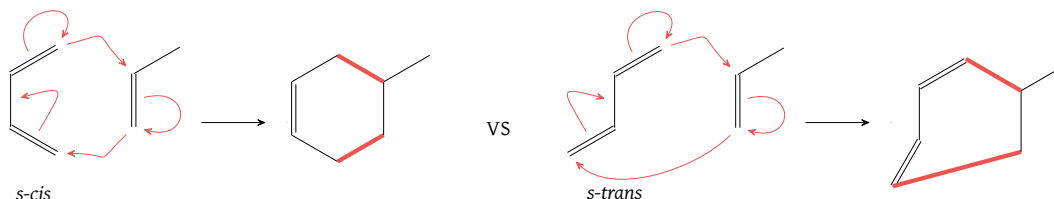


Figure 14: s-cis (left) versus s-trans (right) diene orientation; the s-trans orientation induces substantial ring strain; a ring must contain at least eight atoms to accommodate a trans π bond, and consequently the s-trans reaction depicted above does not proceed

Consider the following dienes as illustrative examples.

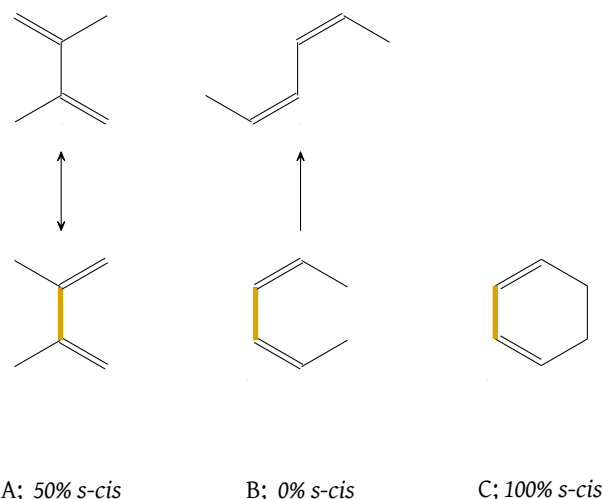


Figure 15: Differences in s-cis orientation prevalence across the three dienes; Diels-Alder reaction rates favor diene C the most and diene B the least

The relative Diels-Alder reaction rates follow the order $C > A > B$. Diene C reacts fastest because its cyclic structure restricts rotation about the relevant single bond, thereby preventing any rearrangement into an s-trans conformation. Dienes A and B, by contrast, can both rotate freely about this single bond; however, B favors the s-trans arrangement, since its s-cis conformation experiences comparatively greater steric hindrance between the alkenyl substituents flanking the relevant single bond.

Core Concept 1.6: Secondary Orbital Overlap

Under certain circumstances, orbitals not directly involved in new bond formation between the diene and dienophile can undergo secondary overlap, thereby lowering the overall E_a . Consider the following Diels-Alder interaction between two cyclopentenes.

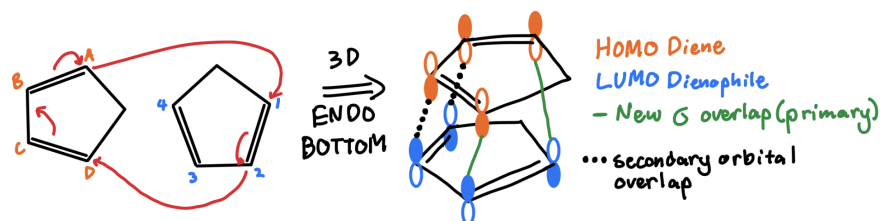


Figure 16: Secondary orbital overlap during an endo Diels-Alder reaction

It is important to note that although favorable mixing of secondary orbital overlaps within the transition state helps stabilize that transition state, steric factors nonetheless disfavor the endo approach. Consequently, ambiguous diene-dienophile pairs typically require experimental investigation to determine which effect predominates in a given case.

Core Concept 1.7: Electronic Effects

Diels-Alder reactions proceed more rapidly when the energies of the interacting orbitals are closer to one another. In other words, the smaller the HOMO-LUMO gap between the reacting molecules, the faster the

reaction rate. This condition is typically achieved through increased conjugation.

However, while increased conjugation typically narrows the HOMO-LUMO gap and accelerates the reaction, it also alters the orbital's wave topology (*i.e.*, the arrangement of its nodes). If the orbital possessing the ideal energy gap happens to place a node on an essential bonding atom, that orbital becomes unusable for the reaction, forcing the system to proceed via a higher-energy, slower pathway instead. Symmetry and orbital overlap must therefore always be satisfied before energy gap considerations become relevant.

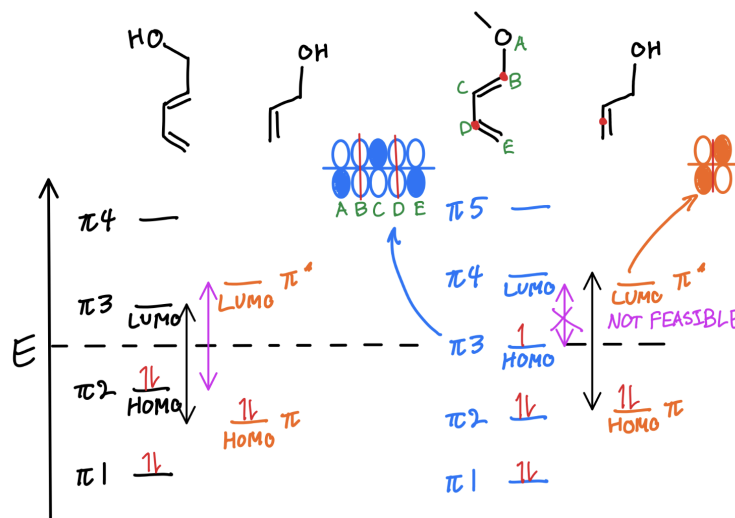


Figure 17: Diels-Alder reaction electronic effects; the reagents on the left actually react faster despite the smaller π_3 - π^* gap of the reagents on the right, because the latter possesses a node on the atom required to form a new σ bond

Core Concept 1.8: Groundstate Destabilization

Given that two sets of similar reagents will produce similar transition states leading to the **same** product, the reactant pair whose diene and dienophile begin at a higher ground-state energy level (*due to a factor such as steric hindrance*) will exhibit a faster Diels-Alder reaction rate. Consider the following diene-dienophile pairs, both of which lead to the same product.

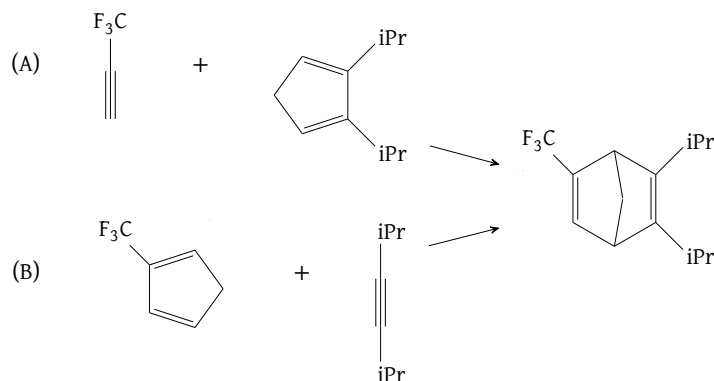


Figure 18: caption

In Figure 18, diene-dienophile pair A possesses a greater degree of steric hindrance between the isopropyl

substituents within the diene, placing it at a higher energy level relative to the pair B.

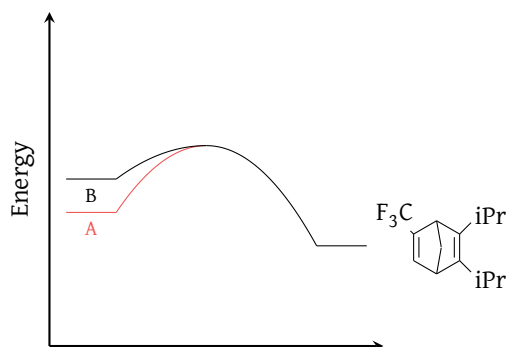


Figure 19: Energy diagrams for the corresponding diene-dienophile pairs from Figure 18; the E_a for compound pair B is substantially smaller than that of pair A, owing to the former's relatively higher starting energy state

As shown in the energy diagram above, the E_a is substantially lower for the more unstable reactant pair A, which consequently proceeds at a much faster rate, since its overall ΔH is lower.